AGENT 3: Printer Module, Batch Printing, Audit & Deployment (Phases 6–8)

You are completing the GS1 DataMatrix UDI Label Management System. Phases 1–5 are done. You are responsible for Phases 6–8: printer connectivity, batch/sequential printing, audit trail UI, and deployment.

Read MASTER-INSTRUCTIONS.md and review the existing codebase first.

CRITICAL LABEL FORMAT RULE

LOT MODE: (01) GTIN + (10) Lot + (11) Mfg. NO expiration. Batch = N identical copies. **SERIAL MODE:** (01) GTIN + (17) Exp + (11) Mfg + (21) Serial. WITH expiration. Batch = sequential serial numbers.

EXISTING CODEBASE

- Complete frontend with product selector (39 products), auto-generated lots, auto-suggested serials
- Label template designer with react-konva, text alignment support
- ZPL generation (lot/serial aware), PDF/PNG export
- Serial counter service tracking last-used serial per prefix family per year
- All JSON stores operational

PHASE 6: PRINTER CONNECTION MODULE

Same as original Agent 3 spec (Steps 6.1–6.5): Zebra Browser Print client, Network TCP driver, Printer API, Printer Manager UI.

No changes from original spec.

PHASE 7: BATCH / SEQUENTIAL PRINTING

Step 7.1 — Print Job API

`POST /api/print-jobs` — Create and process print job:

1. Validate inputs (Zod)

2. Load template from templateStore
3. **LOT MODE:** generate ZPL with `generateLotBatchZpl()`, total = copiesPerLabel. No serials.
4. **SERIAL MODE with batch:** a. Read serial counter for the prefix family + year b. Check that requested range doesn't overlap with already-used serials c. Generate serials, then `generateBatchZpl()` d. **After successful print: update serial counter** with highest serial used e. Store serial range in print job record for audit
5. **SERIAL MODE single:** same as batch but range of 1
6. `printJobStore.append(newJob)` with status
7. For network printers: send via NetworkTcpDriver
8. For Zebra Browser Print: return ZPL for client-side delivery
9. Log to audit

Serial counter update is CRITICAL: The counter must only update AFTER labels are confirmed printed, not when the job is created. If a job fails, the counter stays unchanged and those serials can be reused.

Step 7.2 — Batch Print UI

`BatchPrintPage.tsx`: Guided workflow.

Step 1: Product & Data

- Product already selected on label creator page
- Shows GTIN, mode, lot/serial info

Step 2: Quantity / Batch Config

- **LOT MODE** → "Number of copies" input (N identical labels)
- **SERIAL MODE** → toggle: "Single Label" or "Batch Print"
  - Single: serial number input (auto-suggested from counter)
  - Batch: prefix (from product, read-only), start (editable, auto-suggested counter+1), end (editable), step
  - **Both start and end are editable** — auto-fill is a suggestion, not a lock
  - Serial preview shows first 20 + last 5

Step 3: Print Settings

- Printer selector, copies per label

Step 4: Confirmation

- Full summary
- LOT: shows lot number, copies, NO expiration
- SERIAL: shows serial range, expiration date
- "I confirm" checkbox → enables Print button

Step 5: Printing

- Progress bar, pause/resume/cancel
- Client sends ZPL chunks to Zebra Browser Print
- Updates server after each chunk
- **On SERIAL complete:** PATCH serial counter with highest printed serial

Step 7.3 — Print History

Same as original spec. Additional fields visible:

- For serial jobs: show prefix family, serial range start–end
- For lot jobs: show lot number, copy count
- Product name shown in history (from productId)

Step 7.4 — Test Phase 7

Same as original spec, plus:

1. Print serial batch BWIII2026-3000 to 3004 → serial counter updates to 3004
2. Next serial batch auto-suggests start=3005
3. Attempt to print overlapping range (3002–3006) → warning about already-used serials
4. Failed print job → serial counter NOT updated
5. LOT batch → serial counter unchanged

PHASE 8: AUDIT, POLISH & DEPLOYMENT

Step 8.1 — Audit Log Viewer

Same as original spec. AuditLogViewer.tsx: admin-only, read-only table with filters and CSV export.

Step 8.2 — Navigation

```
/login              → LoginPage
/                   → redirect to /labels
/labels             → LabelDataForm (with product selector, preview, print/export)
/templates          → TemplateList
/templates/designer → LabelDesigner
/products           → ProductList
/print-history      → PrintHistory
/print-history/:id  → PrintJobDetail
/admin/users        → UserManagement
/admin/printers     → PrinterManager
/admin/audit        → AuditLogViewer
```

Step 8.3 — UX Polish

Same as original spec: loading spinners, toast notifications, empty states, unsaved changes warning, session timeout.

Step 8.4 — LAN Deployment

Same as original spec: bind to 0.0.0.0, print queue for LAN users, Windows Firewall instructions.

Step 8.5 — Build Script

Same as original spec.

Step 8.6 — README

Same as original spec, plus:

- Document the 39 pre-seeded products
- Explain serial counter behavior
- Explain lot auto-generation

Step 8.7 — Final Testing

Same as original spec, plus:

1. Select each of the 39 products → correct mode, GTIN, prefix
2. Serial counter persists across server restarts
3. Serial counter increments correctly across multiple batch prints
4. Lot auto-generation works for all 30 lot-mode products

5. Products page shows all 39 products grouped correctly